

15.10 Solution: This problem can be solved in the same way as Problem 14.22, and the difference is that, there is no fundamental frequency for non-periodic motion.

(a) Consider the Fourier transform of the velocity. Similar to Problem 14.22,

$$\begin{aligned} \mathbf{v}(\omega) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{i\omega t} \dot{\mathbf{r}}(t) dt \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \exp \left\{ \frac{i\omega}{\omega_0} (\xi + \epsilon \sinh \xi) \right\} \begin{pmatrix} a \sinh \xi \\ -b \cosh \xi \end{pmatrix} d\xi. \end{aligned}$$

Make the integration variable change, $\xi \rightarrow \xi + i\pi/2$, and with $\sinh \xi \rightarrow i \cosh \xi$, the x -component can be evaluated as

$$v_x(\omega) = \frac{iae^{-\omega\pi/2\omega_0}}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \exp \left\{ \frac{i\omega}{\omega_0} \xi - \frac{\omega}{\omega_0} \epsilon \cosh \xi \right\} \cosh \xi d\xi.$$

Using the following identity,

$$K_\nu(z) = \int_0^{+\infty} e^{-z \cosh t} \cosh(\nu t) dt,$$

we can express the above integral as

$$\begin{aligned} v_x(\omega) &= \frac{iae^{-\omega\pi/2\omega_0}}{2\sqrt{2\pi}} \int_{-\infty}^{+\infty} \left(\exp \left\{ \left(\frac{i\omega}{\omega_0} + 1 \right) \xi - \frac{\omega}{\omega_0} \epsilon \cosh \xi \right\} \right. \\ &\quad \left. + \exp \left\{ \left(\frac{i\omega}{\omega_0} - 1 \right) \xi - \frac{\omega}{\omega_0} \epsilon \cosh \xi \right\} \right) d\xi \\ &= \frac{iae^{-\omega\pi/2\omega_0}}{\sqrt{2\pi}} \int_0^{+\infty} \left(\exp \left\{ -\frac{\omega}{\omega_0} \epsilon \cosh \xi \right\} \cosh \left(\left(\frac{i\omega}{\omega_0} + 1 \right) \xi \right) \right. \\ &\quad \left. + \exp \left\{ -\frac{\omega}{\omega_0} \epsilon \cosh \xi \right\} \cosh \left(\left(\frac{i\omega}{\omega_0} - 1 \right) \xi \right) \right) d\xi \\ &= \frac{iae^{-\omega\pi/2\omega_0}}{\sqrt{2\pi}} \left[K_{i\omega/\omega_0+1} \left(\frac{\omega\epsilon}{\omega_0} \right) + K_{i\omega/\omega_0-1} \left(\frac{\omega\epsilon}{\omega_0} \right) \right]. \end{aligned}$$

For modified Bessel function, we have the recurrence relation

$$K_{\nu+1}(z) + K_{\nu-1}(z) = 2K'_\nu(z),$$

and

$$v_x(\omega) = \frac{2iae^{-\omega\pi/2\omega_0}}{\sqrt{2\pi}} K'_{i\omega/\omega_0} \left(\frac{\omega\epsilon}{\omega_0} \right).$$

Therefore, the Fourier transform of $x(t)$ is

$$x(\omega) = \frac{i}{\omega} v_x(\omega) = -\frac{2ae^{-\omega\pi/2\omega_0}}{\sqrt{2\pi}\omega} K'_{i\omega/\omega_0} \left(\frac{\omega\epsilon}{\omega_0} \right).$$

Similarly, for the y -component,

$$v_y(\omega) = -\frac{ibe^{-\omega\pi/2\omega_0}}{\sqrt{2\pi}} \left[K_{i\omega/\omega_0+1} \left(\frac{\omega\epsilon}{\omega_0} \right) - K_{i\omega/\omega_0-1} \left(\frac{\omega\epsilon}{\omega_0} \right) \right].$$

Using the recurrence relation

$$K_{\nu+1}(z) - K_{\nu-1}(z) = \frac{2\nu}{z} K_{\nu}(z),$$

we have

$$v_y(\omega) = \frac{2be^{-\omega\pi/2\omega_0}}{\sqrt{2\pi\epsilon}} K_{i\omega/\omega_0} \left(\frac{\omega\epsilon}{\omega_0} \right),$$

and

$$y(\omega) = \frac{i}{\omega} v_y(\omega) = \frac{2ibe^{-\omega\pi/2\omega_0}}{\sqrt{2\pi\omega\epsilon}} K_{i\omega/\omega_0} \left(\frac{\omega\epsilon}{\omega_0} \right).$$

Then, from the dipole approximation,

$$\frac{dI}{d\omega} I(\omega, b) = \frac{4}{3c^3} |\ddot{\mathbf{d}}(\omega)|^2 = \frac{4\omega^4}{3c^3} |\mathbf{d}(\omega)|^2,$$

where $\mathbf{d}(\omega) = ze(x(\omega), y(\omega))$, we will have the final result

$$\begin{aligned} \frac{d}{d\omega} I(\omega, b) &= \frac{4\omega^4}{3c^3} \left[\frac{2a^2 z^2 e^2 e^{-\pi\omega/\omega_0}}{\pi\omega^2} \left[K'_{i\omega/\omega_0} \left(\frac{\omega\epsilon}{\omega_0} \right) \right]^2 + \frac{2b^2 z^2 e^2 e^{-\pi\omega/\omega_0}}{\pi\omega^2 \epsilon^2} \left[K_{i\omega/\omega_0} \left(\frac{\omega\epsilon}{\omega_0} \right) \right]^2 \right] \\ &= \frac{8}{3\pi} \frac{(ze a \omega)^2}{c^3} e^{-\pi\omega/\omega_0} \left[\left[K'_{i\omega/\omega_0} \left(\frac{\omega\epsilon}{\omega_0} \right) \right]^2 + \frac{b^2/a^2}{\epsilon^2} \left[K_{i\omega/\omega_0} \left(\frac{\omega\epsilon}{\omega_0} \right) \right]^2 \right] \\ &= \frac{8}{3\pi} \frac{(ze a \omega)^2}{c^3} e^{-\pi\omega/\omega_0} \left[\left[K'_{i\omega/\omega_0} \left(\frac{\omega\epsilon}{\omega_0} \right) \right]^2 + \frac{\epsilon^2 - 1}{\epsilon^2} \left[K_{i\omega/\omega_0} \left(\frac{\omega\epsilon}{\omega_0} \right) \right]^2 \right]. \end{aligned}$$

(b) From part (a), the radiation cross section is

$$\begin{aligned} \frac{d\chi(\omega)}{d\omega} &= 2\pi \int_0^{+\infty} \frac{d}{d\omega} I(\omega, b) b db \\ &= \frac{16}{3} \frac{(ze a \omega)^2}{c^3} e^{-\pi\omega/\omega_0} \int_0^{+\infty} \left[\left[K'_{i\omega/\omega_0} \left(\frac{\omega\epsilon}{\omega_0} \right) \right]^2 + \frac{\epsilon^2 - 1}{\epsilon^2} \left[K_{i\omega/\omega_0} \left(\frac{\omega\epsilon}{\omega_0} \right) \right]^2 \right] b db. \end{aligned}$$

Since $\epsilon = \sqrt{1 + (b/a)^2}$, then $b = a\sqrt{\epsilon^2 - 1}$, $db = a\epsilon d\epsilon/\sqrt{\epsilon^2 - 1}$, and $b db = a^2 \epsilon d\epsilon$. The integral becomes

$$a^2 \int_1^{+\infty} \left[\left[K'_{i\omega/\omega_0} \left(\frac{\omega\epsilon}{\omega_0} \right) \right]^2 + \frac{\epsilon^2 - 1}{\epsilon^2} \left[K_{i\omega/\omega_0} \left(\frac{\omega\epsilon}{\omega_0} \right) \right]^2 \right] \epsilon d\epsilon,$$

which, upon another variable change, $\epsilon = (\omega_0/\omega)x$, becomes

$$a^2 \left(\frac{\omega_0}{\omega} \right)^2 \int_{\omega/\omega_0}^{+\infty} \left[\left[K'_{i\omega/\omega_0}(x) \right]^2 + \left(1 - \frac{\omega^2/\omega_0^2}{x^2} \right) \left[K_{i\omega/\omega_0}(x) \right]^2 \right] x dx.$$

The first term can be evaluated with integration by parts,

$$\begin{aligned} \int_{\omega/\omega_0}^{+\infty} x \left[K'_{i\omega/\omega_0}(x) \right]^2 dx &= \int_{\omega/\omega_0}^{+\infty} x K'_{i\omega/\omega_0}(x) d(K_{i\omega/\omega_0}(x)) \\ &= x K_{i\omega/\omega_0}(x) K'_{i\omega/\omega_0}(x) \Big|_{\omega/\omega_0}^{+\infty} - \int_{\omega/\omega_0}^{+\infty} K_{i\omega/\omega_0}(x) d(x K'_{i\omega/\omega_0}(x)). \end{aligned}$$

The modified Bessel function of the second kind satisfies the following ODE,

$$\frac{d}{dx} [xK'_\nu(x)] = x \left(1 + \frac{\nu^2}{x^2}\right) K_\nu(x),$$

then we are left with the boundary term only,

$$\frac{d\chi(\omega)}{d\omega} = \frac{16}{3} \frac{(ze a \omega)^2}{c^3} e^{-\pi\omega/\omega_0} a^2 \left(\frac{\omega_0}{\omega}\right)^2 \left[xK_{i\omega/\omega_0}(x) K'_{i\omega/\omega_0}(x) \right] \Big|_{\omega/\omega_0}^{+\infty}.$$

Since $K_\nu(x)$ vanishes exponentially at infinity, and notice that $\omega_0 = v/a$, we are finally left with

$$\frac{d\chi(\omega)}{d\omega} = \frac{16}{3} \frac{(zeav)^2}{c^3} e^{-\pi\omega/\omega_0} \frac{\omega}{\omega_0} K_{i\omega/\omega_0} \left(\frac{\omega}{\omega_0}\right) \left[-K'_{i\omega/\omega_0} \left(\frac{\omega}{\omega_0}\right) \right].$$

(c) For $\omega/\omega_0 \ll 1$, using the asymptotic form of the Bessel function,

$$K_\nu(x) \rightarrow -\log\left(\frac{x}{2}\right), \quad K'_\nu(x) \rightarrow -\frac{1}{x},$$

we can write the cross section as

$$\frac{d\chi(\omega)}{d\omega} = \frac{16}{3} \frac{(zeav)^2}{c^3} \frac{\omega}{\omega_0} \left[-\log\left(\frac{\omega}{2\omega_0}\right) \right] \left(\frac{\omega_0}{\omega}\right) = \frac{16}{3} \frac{(zeav)^2}{c^3} \log\left(\frac{2\omega_0}{\omega}\right).$$

Since $a = Ze^2/mv^2$, we then have

$$\begin{aligned} \frac{d\chi(\omega)}{d\omega} &= \frac{16}{3} \frac{(Zze^3/mv)^2}{c^3} \log\left(\frac{2mv^3}{Zze^2\omega}\right) \\ &= \frac{16}{3} \frac{Z^2e^2}{c^3} \left(\frac{z^2e^2}{m}\right)^2 \frac{1}{(\beta c)^2} \log\left(\frac{2mv^3}{Zze^2\omega}\right) \\ &= \frac{16}{3} \frac{Z^2e^2}{c} \left(\frac{z^2e^2}{mc^2}\right)^2 \frac{1}{\beta^2} \log\left(\frac{2mv^3}{Zze^2\omega}\right), \end{aligned}$$

which agrees with Eq. (15.25), up to some constant factor.

For $\omega/\omega_0 \gg 1$, using the asymptotic result $K_\nu(x) \rightarrow e^{-x} \sqrt{\pi/2x}$, we can see that the limiting form is proportional to $e^{-(\pi+2)\omega/\omega_0}$.

(d) For attractive Coulomb interaction, the trajectory can be parameterized as

$$x = a(\epsilon - \cosh \xi), \quad y = -b \sinh \xi, \quad \omega_0 t = \xi - \epsilon \sinh \xi,$$

then we can repeat the calculation as in part (a),

$$\mathbf{v}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \exp \left\{ \frac{i\omega}{\omega_0} (\xi - \epsilon \sinh \xi) \right\} \begin{pmatrix} -a \sinh \xi \\ -b \cosh \xi \end{pmatrix} d\xi.$$

This time, make the change of integration variable as $\xi \rightarrow \xi - i\pi/2$, then $\sinh \xi \rightarrow -i \cosh \xi$ and $\cosh \xi \rightarrow -i \sinh \xi$, and

$$\mathbf{v}(\omega) = \frac{ie^{\omega\pi/2\omega_0}}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \exp \left\{ \frac{i\omega}{\omega_0} \xi - \frac{\omega}{\omega_0} \epsilon \cosh \xi \right\} \begin{pmatrix} a \cosh \xi \\ b \sinh \xi \end{pmatrix} d\xi.$$

This is familiar from part (a) and we can write down the result directly as

$$\mathbf{v}(\omega) = \frac{ie^{\omega\pi/2\omega_0}}{\sqrt{2\pi}} \begin{pmatrix} a \left[K_{i\omega/\omega_0+1} \left(\frac{\omega\epsilon}{\omega_0} \right) + K_{i\omega/\omega_0-1} \left(\frac{\omega\epsilon}{\omega_0} \right) \right] \\ b \left[K_{i\omega/\omega_0+1} \left(\frac{\omega\epsilon}{\omega_0} \right) - K_{i\omega/\omega_0-1} \left(\frac{\omega\epsilon}{\omega_0} \right) \right] \end{pmatrix}.$$

Now, using the relation

$$K_\nu(x) = \frac{\pi}{2} e^{i(\nu+1)\pi/2} H_\nu^{(1)}(ix),$$

we can express the above result in terms of Hankel functions,

$$\begin{aligned} & K_{i\omega/\omega_0+1} \left(\frac{\omega\epsilon}{\omega_0} \right) \pm K_{i\omega/\omega_0-1} \left(\frac{\omega\epsilon}{\omega_0} \right) \\ = & \frac{\pi}{2} \left[\exp \left(i \left(i \frac{\omega}{\omega_0} + 2 \right) \frac{\pi}{2} \right) H_{i\omega/\omega_0+1}^{(1)} \left(i \frac{\omega\epsilon}{\omega_0} \right) \pm \exp \left(i \left(i \frac{\omega}{\omega_0} \right) \frac{\pi}{2} \right) H_{i\omega/\omega_0-1}^{(1)} \left(i \frac{\omega\epsilon}{\omega_0} \right) \right] \\ = & -\frac{\pi}{2} e^{-\omega\pi/2\omega_0} \left[H_{i\omega/\omega_0+1}^{(1)} \left(i \frac{\omega\epsilon}{\omega_0} \right) \mp H_{i\omega/\omega_0-1}^{(1)} \left(i \frac{\omega\epsilon}{\omega_0} \right) \right], \end{aligned}$$

and

$$\mathbf{v}(\omega) = -i\sqrt{\frac{\pi}{8}} \begin{pmatrix} a \left[H_{i\omega/\omega_0+1}^{(1)} \left(i \frac{\omega\epsilon}{\omega_0} \right) - H_{i\omega/\omega_0-1}^{(1)} \left(i \frac{\omega\epsilon}{\omega_0} \right) \right] \\ b \left[H_{i\omega/\omega_0+1}^{(1)} \left(i \frac{\omega\epsilon}{\omega_0} \right) + H_{i\omega/\omega_0-1}^{(1)} \left(i \frac{\omega\epsilon}{\omega_0} \right) \right] \end{pmatrix}.$$

Finally, from the recurrence relation, we know

$$H_{\nu-1}^{(1)}(x) - H_{\nu+1}^{(1)}(x) = 2H_\nu^{(1)'}(x), \quad H_{\nu-1}^{(1)}(x) + H_{\nu+1}^{(1)}(x) = \frac{2\nu}{x} H_\nu^{(1)}(x),$$

and the Fourier transform of the velocity becomes

$$\mathbf{v}(\omega) = -i\sqrt{\frac{\pi}{8}} \begin{pmatrix} -2aH_{i\omega/\omega_0}^{(1)'} \left(i \frac{\omega\epsilon}{\omega_0} \right) \\ \frac{2b}{\epsilon} H_{i\omega/\omega_0}^{(1)} \left(i \frac{\omega\epsilon}{\omega_0} \right) \end{pmatrix},$$

with

$$\begin{aligned} \frac{d}{d\omega} I(\omega, b) &= \frac{4\omega^4}{3c^3} \cdot (ze)^2 \cdot \frac{\pi}{8\omega^2} \left\{ 4a^2 \left[H_{i\omega/\omega_0}^{(1)'} \left(i \frac{\omega\epsilon}{\omega_0} \right) \right]^2 - \frac{4b^2}{\epsilon^2} \left[H_{i\omega/\omega_0}^{(1)} \left(i \frac{\omega\epsilon}{\omega_0} \right) \right]^2 \right\} \\ &= \frac{2\pi}{3} \frac{(ze a \omega)^2}{c^3} \left\{ \left[H_{i\omega/\omega_0}^{(1)'} \left(i \frac{\omega\epsilon}{\omega_0} \right) \right]^2 - \frac{\epsilon^2 - 1}{\epsilon^2} \left[H_{i\omega/\omega_0}^{(1)} \left(i \frac{\omega\epsilon}{\omega_0} \right) \right]^2 \right\}, \end{aligned}$$

since $H_{i\omega/\omega_0}^{(1)}(i\omega\epsilon/\omega_0)$ is pure imaginary.

To calculate the radiation cross section, consider the integral

$$\int_0^{+\infty} \left\{ \left[H_{i\omega/\omega_0}^{(1)'} \left(i \frac{\omega\epsilon}{\omega_0} \right) \right]^2 - \frac{\epsilon^2 - 1}{\epsilon^2} \left[H_{i\omega/\omega_0}^{(1)} \left(i \frac{\omega\epsilon}{\omega_0} \right) \right]^2 \right\} b db$$

$$= a^2 \left(\frac{\omega_0}{\omega} \right)^2 \int_{\omega/\omega_0}^{+\infty} \left\{ \left[H_{i\omega/\omega_0}^{(1)'}(ix) \right]^2 - \left(1 - \frac{\omega^2/\omega_0^2}{x^2} \right) \left[H_{i\omega/\omega_0}^{(1)}(ix) \right]^2 \right\} x dx. \quad (1)$$

By integration by parts, as in part (b), the first integral becomes

$$\begin{aligned} & \int_{\omega/\omega_0}^{+\infty} x \left[H_{i\omega/\omega_0}^{(1)'}(ix) \right]^2 dx \\ = & - \int_{i\omega/\omega_0}^{+i\infty} x \left[H_{i\omega/\omega_0}^{(1)'}(x) \right]^2 dx = - \int_{i\omega/\omega_0}^{+i\infty} x H_{i\omega/\omega_0}^{(1)'}(x) d \left[H_{i\omega/\omega_0}^{(1)}(x) \right] \\ = & -x H_{i\omega/\omega_0}^{(1)}(x) H_{i\omega/\omega_0}^{(1)'}(x) \Big|_{i\omega/\omega_0}^{+i\infty} + \int_{i\omega/\omega_0}^{+i\infty} H_{i\omega/\omega_0}^{(1)}(x) d \left[x H_{i\omega/\omega_0}^{(1)'}(x) \right]. \end{aligned} \quad (2)$$

The Hankel function satisfies the Bessel ODE,

$$\frac{d}{dx} [x H_{\nu}^{(1)'}(x)] = -x \left(1 - \frac{\nu^2}{x^2} \right) H_{\nu}^{(1)}(x),$$

which means that the second term in Eq. (2) becomes

$$- \int_{i\omega/\omega_0}^{+i\infty} x \left(1 + \frac{\omega^2/\omega_0^2}{x^2} \right) \left[x H_{i\omega/\omega_0}^{(1)}(x) \right]^2 dx = \int_{\omega/\omega_0}^{+\infty} x \left(1 - \frac{\omega^2/\omega_0^2}{x^2} \right) \left[H_{i\omega/\omega_0}^{(1)}(ix) \right]^2 dx,$$

which exactly cancels the second term of Eq. (1). Therefore, we are only left with the boundary term, which gives

$$\begin{aligned} \frac{d}{d\omega} \chi(\omega) &= \frac{4\pi^2}{3} \frac{(zeav)^2}{c^3} a^2 \left(\frac{\omega_0}{\omega} \right)^2 \left[-ix H_{i\omega/\omega_0}^{(1)}(ix) H_{i\omega/\omega_0}^{(1)'}(ix) \right] \Big|_{\omega/\omega_0}^{+\infty} \\ &= \frac{4\pi^2}{3} \frac{(zeav)^2}{c^3} \frac{i\omega}{\omega_0} H_{i\omega/\omega_0}^{(1)} \left(\frac{i\omega}{\omega_0} \right) H_{i\omega/\omega_0}^{(1)'} \left(\frac{i\omega}{\omega_0} \right). \end{aligned}$$

Recall that $H_{i\omega/\omega_0}^{(1)}(i\omega\epsilon/\omega_0)$ is pure imaginary and $H_{i\omega/\omega_0}^{(1)'}(i\omega\epsilon/\omega_0)$ is real, the radiation cross section is guaranteed to be real.

For $\omega/\omega_0 \ll 1$, we have

$$H_{i\omega/\omega_0}^{(1)} \left(\frac{i\omega}{\omega_0} \right) \sim H_0^{(1)} \left(\frac{i\omega}{\omega_0} \right) \sim \frac{2}{\pi i} \log \left(\frac{2\omega_0}{\gamma\omega} \right),$$

and

$$H_{i\omega/\omega_0}^{(1)'} \left(\frac{i\omega}{\omega_0} \right) \sim H_0^{(1)'} \left(\frac{i\omega}{\omega_0} \right) \sim \frac{2\omega_0}{\pi\omega}.$$

Therefore, in this limit, the radiation cross section is reduced to

$$\frac{d}{d\omega} \chi(\omega) = \frac{4\pi^2}{3} \frac{(zeav)^2}{c^3} \cdot \frac{i\omega}{\omega_0} \left[\frac{2}{\pi i} \log \left(\frac{2\omega_0}{\gamma\omega} \right) \right] \left(\frac{2\omega_0}{\pi\omega} \right) = \frac{16}{3} \frac{(zeav)^2}{c^3} \log \left(\frac{2\omega_0}{\gamma\omega} \right),$$

which also agrees with Eq. (15.25).

For $\omega/\omega_0 \gg 1$, it can be shown (see Landau and Lifshitz, *The Classical Theory of fields*, Section 70), the radiation cross section is frequency independent.